

EBS VOLUME MANAGEMENT AND BACKUP STRATEGIES

❖ What is Amazon EBS?

- Amazon **EBS** (Elastic Block Store) is a **block-level storage service** provided by **AWS** that is used to store persistent data for Amazon EC2 (Elastic Compute Cloud) instances.
- EBS volumes behave like traditional **hard drives** or **SSDs** attached to **virtual machines**, and they persist independently of the EC2 instance's lifecycle.
- EBS is designed to provide high-performance, scalable, and durable storage that can be attached to a single EC2 instance at a time (though multi-attach is possible with certain volume types).
- It supports both **boot volumes** and **data volumes**, and you can take **snapshots** for backup, recovery, and replication.
- **EBS Key Features**
 1. **Persistent Storage**
 - *Data remains available even after the EC2 instance is stopped or terminated.*
 2. **Performance-Optimized**
 - *Offers SSD and HDD-based storage options tailored for transactional or throughput-intensive workloads.*
 3. **Scalable**
 - *EBS volumes can be resized and their performance can be adjusted without downtime.*
 4. **Snapshots and Backup**
 - *EBS allows point-in-time snapshots that are stored in Amazon S3 and can be used to restore volumes or create new ones.*
 5. **Encryption**
 - *Supports encryption at rest and in transit using AWS Key Management Service (KMS).*
 6. **Availability and Durability**
 - *EBS volumes are automatically replicated within the same Availability Zone to prevent data loss due to hardware failure.*
 7. **Attach and Detach Volumes**
 - *Volumes can be attached to or detached from instances as needed.*
 8. **Integration with Other AWS Services**
 - *Works seamlessly with EC2, CloudWatch, IAM, Auto Scaling, and more.*

❖ Types of EBS Volumes

- **Volume types**
 1. Solid state drive (SSD) volumes
 2. Hard disk drive (HDD) volumes
 3. Previous generation volumes

Uses of Each volume type see on Day 6 EBS Volume

https://edublitz.edmingle.com/v3/myaccount/course/146217/lesson/1479706?lesson=1479706§ion=351121&class_id=352537

❖ What is an EBS Snapshot?

- An EBS Snapshot is a **point-in-time backup** of an Amazon Elastic Block Store (**EBS**) volume.
- Snapshots **allow you to** preserve the data on an EBS volume by **creating a backup that is stored** in Amazon S3.
- Although the **snapshot data is stored in S3**, it is not visible or accessible through the S3 console because AWS manages it **internally**.

➤ **Snapshots can be used to:**

- **Restore Data** :- Create a new EBS volume from a snapshot to recover lost or corrupted data.
- **Create Copies in Other AZs or Regions** :- Snapshots can be copied to other AWS Regions for **disaster recovery** or **geographic redundancy**.
- **Launch New Instances** :- Use a snapshot to create a volume, then attach it to a new EC2 instance — useful for **cloning environments**.
- **Automate Backups** :- Schedule regular snapshots using **Snapshot Lifecycle Policies** to automate backup and retention.
- **Migrate Data** :- Move EBS volumes between instances, availability zones, or regions using snapshots.
- **Create AMIs (Amazon Machine Images)** :- Snapshots of root volumes are used to build custom AMIs for launching EC2 instances.

❖ **When Should You Use EBS Snapshots?**

- Before Performing System Changes or Updates
- Regular Backups
- Disaster Recovery and Compliance
- Migrating Volumes Across Regions
- Cloning or Creating New Environments

❖ **What is Amazon Data Lifecycle Manager (DLM)?**

- Amazon Data Lifecycle Manager (DLM) is a feature within the EC2 service in AWS that allows users to automate the **creation**, **retention**, and **deletion** of Amazon EBS snapshots.
- With DLM, system administrators can define lifecycle policies to ensure that snapshots are created on a regular schedule and that older backups are automatically deleted based on retention rules.

❖ **Benefits of Using Snapshot Lifecycle Policies**

- Automated Backup Scheduling
- Consistent Retention Management
- Improved Data Protection
- Cost Optimization
- Tag-Based Flexibility

Example Use Case snap shot

- An organization wants to:
- Automatically back up production volumes daily.
- Retain each daily backup for 7 days.
- Avoid manual backup tasks.

Note: - Retention: 7 snapshots (oldest snapshot automatically deleted when 8th is created)

NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINT

xvda 202:0 0 8G 0 disk /

xvdf 202:80 0 10G 0 disk

Step 1 :- Create a Partition

Use the **fdisk** utility to create a partition on the disk.

```
sudo fdisk /dev/xvdf
```

Step 2 :- Format the Partition

Next, format the new partition with a filesystem. In most Linux systems, ext4 is used.

- `sudo mkfs.ext4 /dev/xvdf1`
- ***mkfs -t ext4 /dev/xvdf***

Step 3 :- Create a Mount Point

You must create a directory where the new partition will be mounted. For example:

```
sudo mkdir /mnt/data
```

You can replace /mnt/data with any directory path as per your need.

Mount the Partition Temporarily

Step 4 :- To make sure everything works correctly, mount the partition manually:

```
sudo mount /dev/xvdf1 /mnt/data
```

You can verify the mount using:

df -h

Step 5 :- Edit /etc/fstab for Permanent Mount

Open the /etc/fstab file in a text editor:

```
sudo nano /etc/fstab
```

Add a new line at the bottom of the file using the UUID retrieved earlier:

```
UUID=e2a4f7a0-9c51-49e6-91f0-631dc06a1423 /mnt/data ext4 defaults,nofail 0 2
```

Commands for practice.

1. ***apt update -y***
2. ***apt install nfs—utils***
3. ***apt-cache search nfs***
4. ***apt install nfs-common***
5. ***vim /etc/fstab***
6. ***usw-est,amazonaws.com:mnt/nfs4 default 0 0***
7. ***mount -a***

[General interview Question]

Q1. What is the difference between EBS Snapshots and AMIs?

- **EBS Snapshot:** A backup of a specific EBS volume that can be used to **restore** or create new volumes.
- **AMI (Amazon Machine Image):** A **pre-configured template** that **includes** an operating system, application software, and EBS snapshots, allowing instances to be launched with **predefined** settings.

Q2. Can you attach an EBS volume to multiple EC2 instances at the same time? If yes, how?

- **Yes**, but only with **EBS Multi-Attach** feature, which is supported for **Provisioned IOPS SSD (io2)** volumes.
- **Multiple EC2 instances** within the **same Availability Zone (AZ)** can simultaneously access a single io2 volume.
- Applications must be designed to **handle concurrent access** (e.g. clustered databases or shared file systems).

Q3. How does EBS handle data consistency for snapshots of running instances?

- When you take a **snapshot** of an EBS volume attached to a **running EC2 instance**, Amazon EBS ensures **crash-consistent** snapshots — but not **application-consistent**, unless you take additional steps.

Example:

1. Crash-Consistent Snapshots (Default Behavior)

- EBS snapshots capture the volume as if the power was suddenly cut.
- All written data is already saved to disk

Q4. What happens when you delete an EBS volume? Can you recover the data?

- If an EBS volume is **deleted** without snapshots, the data is **permanently lost**.
- If snapshots **exist**, a new volume can be **created** from the latest snapshot to **restore** the data.
- AWS **does not provide** soft delete or recycle bin for EBS volumes unless snapshots are taken.

Q5. What is the difference between AWS EBS gp2 and gp3 volumes?

- **gp2 (General Purpose SSD):** Performance scales with **volume size**(3 IOPS/GB with bursting).
- **gp3 (Next-Generation SSD):** Performance (IOPS and throughput) is **decoupled** from **storage size**, allowing manual provisioning of **IOPS** and throughput for **better cost** optimization.

Q6. What is the recommended method to migrate an EBS volume from one region to another?

- **Create** an **EBS Snapshot** of the volume.
- **Copy** the snapshot to the **target region** using the AWS Console or CLI.
- **Create** a new EBS volume from the **copied** snapshot in the **target region**.

Q7. What happens when an EBS volume reaches its maximum IOPS or throughput limit?

- ✓ *When the limits are reached, **disk I/O** operations experience increased latency or are throttled.*
- **To mitigate this:**
 - Upgrade to **Provisioned IOPS SSD (io1/io2)** for higher throughput.
 - Optimize application I/O patterns to **reduce** excessive read/write operations.
 - Enable **Elastic Volumes** to dynamically increase volume size and IOPS.

Q8. How do you prevent data loss in EC2 instances?

- ✓ *Regular EBS Snapshots for backup.*
- Use Auto Recovery to **restart** instances on failure.
- Cross-region **replication** for high availability.
- Instance Store-backed instances should not be used for critical data, as they are **non- persistent**.